Data Catalog Search Assistant: submission summary

Assignment: DAI-041 (AI + Data) · **Candidate:** Nikhil Dubey · **Stack:** Python 3.10+, SQLite (SQL, FTS5), optional Ollama / OpenAI-compatible LLM, Streamlit, pytest

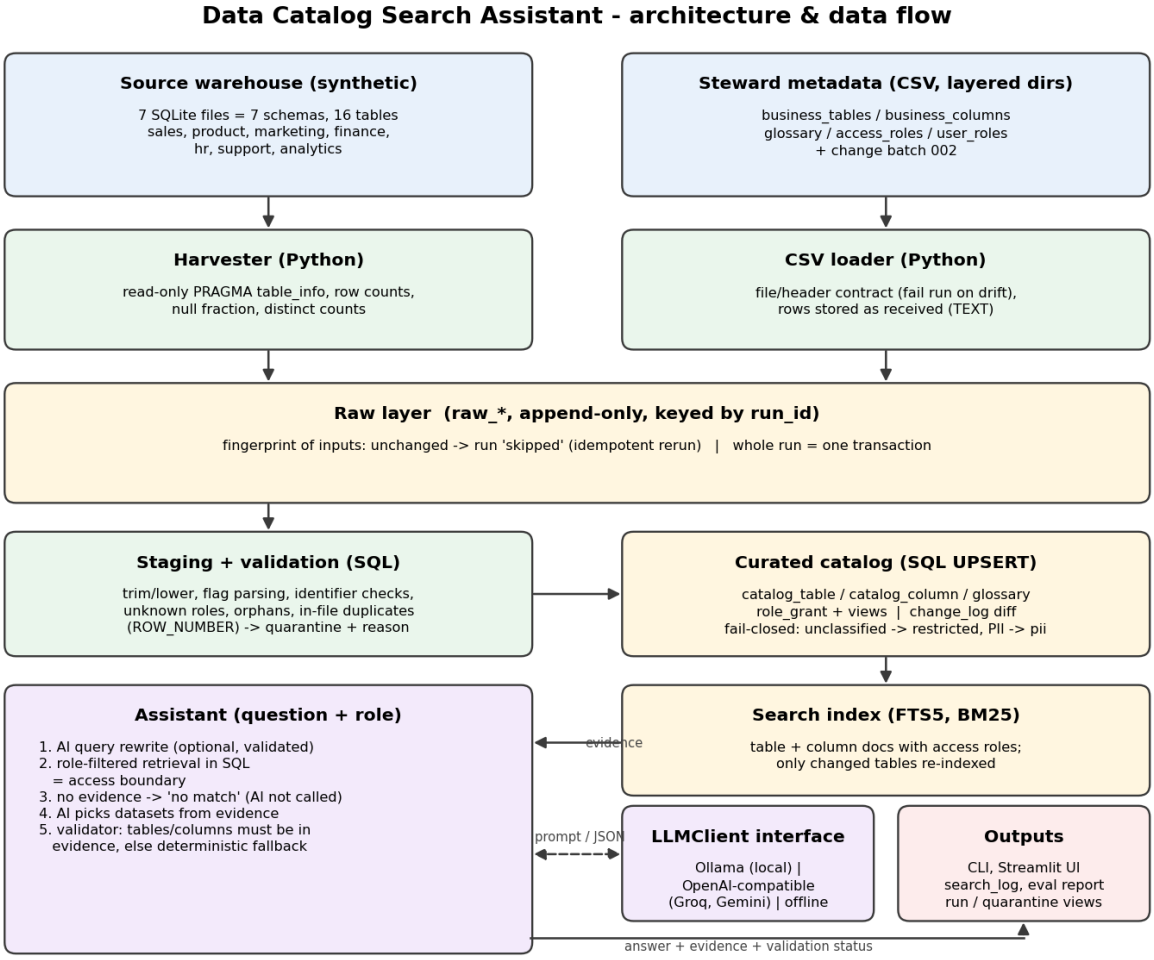
Links: GitHub repository: https://github.com/Nikhil-Dubey-0/data-catalog-search-assistant · Demo video: *to be added*

Problem

Users need to find the right dataset by describing it in plain language. Answers must show **table and column evidence** and respect a simple **access-role** field: nobody should see, or be told about, metadata their role isn't granted.

Solution in one paragraph

A metadata pipeline **harvests** technical metadata (tables, columns, types, null %, distinct counts) from a synthetic 7-schema warehouse and merges it with **steward CSVs** (descriptions, owners, tags, PII flags, access roles, glossary). Rows land in a raw layer unchanged. SQL staging standardises, validates and de-duplicates them, and bad rows go to a **quarantine** table with a reason. Curated tables are maintained with **UPSERT, change log and soft delete**, and only changed tables are re-indexed in an **FTS5** search index. Runs are fingerprinted (identical inputs are skipped) and atomic (one transaction). The assistant retrieves candidates with **BM25 + glossary synonyms**, with the **role filter inside the retrieval SQL**. An optional LLM rewrites the question and explains the best datasets from that evidence, and a **validator rejects any answer citing tables outside the evidence**, falling back to a deterministic answer.



Key decisions

- **Fail closed:** an unclassified table becomes restricted, and an unclassified PII-named column becomes pii.
- **Access enforced at retrieval, not in the prompt:** restricted metadata never reaches the model (tested on the actual prompt).
- **Quarantine, don't drop:** each bad steward row is kept with its reason. A broken file *header* fails the run instead.

- **Glossary instead of embeddings (for now):** transparent and dependency-free, and measured to help (hit@1 0.909 -> 0.955, MRR 0.909 -> 0.977).
- **AI as an untrusted component:** it adds keywords and explanations only. Its output is validated, and the system works fully offline.

Results

Check	Result
Automated tests	49 passed (pipeline, access, search, AI failure modes, HTTP retries, eval gate)
Evaluation (29 labelled questions, offline)	28/29 pass, 0 policy leaks, 0 label violations
Retrieval with glossary	hit@1 0.955, hit@3 1.0, MRR 0.977
Mutation check	disabling the table filter -> 4 tests fail; the column filter -> 1; the validator leak check -> 1
Clean clone + new venv	setup and full test suite pass
Demonstrated edge cases	idempotent rerun, header drift, mid-run rollback, unknown role, hallucinated/leaky/malformed AI output, model outage, schema change

Limitations and next steps

Keyword retrieval misses some paraphrases (eval q24, "bought"), so I would add hybrid BM25 + embeddings. The access model is single-role and the role isn't authenticated, so I would map SSO groups to roles and share policies with the warehouse. SQLite would become Postgres/dbt with a scheduler. The eval set is small and author-labelled; I would grow it and run it in CI per model version. The live LLM path is covered by contract and fake-model tests but wasn't run against a real model in the build environment.

How to run

```
pip install -r requirements.txt
python -m catalog_assistant setup
python -m catalog_assistant ask "employee salaries by department" --role hr_partner
pytest
streamlit run app/streamlit_app.py
```